

CSE210: Floating Point Adder Simulation

Sample Test Cases

Architecture Specification: 16-bit format consisting of 1 Sign bit, 5 Exponent bits (Bias = 15), and 10 Fraction bits. Assumes an implicit leading 1. for normalized numbers.

1. Normal Addition (No Normalization Required)

Tests basic alignment and addition when the result remains within the 1.x format.

- **Calculation:** $12.0 + 0.5 = 12.5$
- **Input 1 (12.0):** 0 10010 1000000000 $(1.100_2 \times 2^3)$
- **Input 2 (0.5):** 0 01110 0000000000 $(1.0_2 \times 2^{-1})$
- **Procedure:** Exponent difference is 4. Shift Input 2 right by 4 and add:
 $1.1000000000_2 + 0.0001000000_2 = 1.1001000000_2$.
- **Expected Output (12.5):** 0 10010 1001000000

2. Addition Requiring Normalization

Tests hardware detection of sums ≥ 2.0 , which require right-shifting the mantissa and incrementing the exponent.

- **Calculation:** $15.0 + 1.0 = 16.0$
- **Input 1 (15.0):** 0 10010 1110000000 $(1.111_2 \times 2^3)$
- **Input 2 (1.0):** 0 01111 0000000000 $(1.0_2 \times 2^0)$
- **Procedure:** Shift Input 2 right by 3. Sum is 10.0000000000_2 . Normalize by shifting right by 1 and incrementing the exponent.
- **Expected Output (16.0):** 0 10011 0000000000

3. Addition Requiring Rounding

Ensures guard, round, and sticky bits properly trigger a round-up when shifted bits fall outside the 10-bit fraction.

- **Calculation:** $16.0 + 0.01171875$
- **Input 1 (16.0):** 0 10011 0000000000 $(1.0_2 \times 2^4)$
- **Input 2 (0.01171875):** 0 01000 1000000000 $(1.1_2 \times 2^{-7})$
- **Procedure:** Shift Input 2 right by 11. The shifted bits represent more than halfway to the least significant bit, forcing a round-up.
- **Expected Output:** 0 10011 0000000001 (Fraction incremented by 1)

4. Overflow Flag Verification

Tests if the hardware flags operations that exceed the maximum representable finite value.

- **Calculation:** Max Positive + Max Positive
- **Input 1:** 0 11110 1111111111 $(1.111..._2 \times 2^{15})$
- **Input 2:** 0 11110 1111111111 $(1.111..._2 \times 2^{15})$
- **Procedure:** The normalized sum pushes the true exponent to 16 (biased 31 or 11111).

- **Expected Output: OVERFLOW FLAG = ON**

5. Underflow Flag Verification

Checks for catastrophic cancellation where the normalized result is too close to zero to maintain a valid exponent.

- **Calculation:** $A - B$ (Subtracting two nearly identical small numbers)
- **Input 1 (A):** 0 01010 0000000001 $(1.0000000001_2 \times 2^{-5})$
- **Input 2 ($-B$):** 1 01010 0000000000 $(-1.0_2 \times 2^{-5})$
- **Procedure:** Subtraction yields $0.0000000001_2 \times 2^{-5}$. Normalizing requires a left shift of 10, dropping the biased exponent to 0 (00000).
- **Expected Output: UNDERFLOW FLAG = ON**